Matched Filter Based on the Radiative Transfer Model for CO_2 Estimation From PRISMA Hyperspectral Data

Nicola Acito, *Member, IEEE*, M. Diani, *Member, IEEE*, M. Alibani, *Member, IEEE*, and G. Corsini, *Life Member, IEEE*

***Abstract*— The rapid growth of hyperspectral satellite missions and the subsequent availability of hyperspectral images have encouraged the remote sensing community to investigate their potential in estimating the concentration of gases, such as methane (CH_4) and carbon dioxide (CO_2), which are related to the greenhouse effect. Though satellite hyperspectral sensors are not specifically designed for this purpose, they are expected to complement more specific satellite missions, such as NASA's Orbiting Carbon Observatory -2 and -3 (OCO-2 and OCO-3), both in terms of enriched temporal sampling and improved spatial resolution. In this work, we present a new method to estimate the column-averaged dry-air mole fraction of CO_2 from hyperspectral data on a per-pixel basis. The method, which is here tailored to PRISMA images, leverages the spectral radiance samples collected in the short-wave InfraRed (SWIR) spectral region around the CO_2 absorption band at 2000 nm. By assuming a linear model to describe the dependence of the observed radiance on the CO_2 concentration, the estimation problem is reduced to matched filtering and can be effectively implemented in compliance with the low computational burden required to perform a pixel-by-pixel analysis. The performance of the presented method is investigated by means of a rigorous, physically based simulator that accurately reproduces the at-sensor radiance allowing one to check the validity of the assumptions and to assess the algorithm accuracy. The results show that the presented algorithm outperforms a benchmark continuum interpolated band ratio (CIBR)-based approach, which has been proposed in the literature to get fast per-pixel estimates of CO_2 concentration.**

***Index Terms*— Atmospheric parameters estimation, carbon dioxide (CO_2) retrieval, satellite hyperspectral data.**

I. Introduction

NATURAL phenomena and anthropogenic activities release carbon dioxide (CO_2) and methane (CH_4) in the atmosphere. CO_2 and CH_4 are the most relevant climate-forcing agents and are closely related to the climate change and global warming [1]. Therefore, their characterization on both a global and local scale is of fundamental importance for monitoring the evolution of climate changes and planning the actions necessary to avoid or limit the harmful effects on the planet.

CO_2 and CH_4 absorb the radiation that propagates in the atmosphere in specific spectral windows located in the short-wave InfraRed (SWIR) portion of the electromagnetic spectrum. The absorption features are directly related to the column-averaged dry-air mole fraction of CO_2 (XCO_2) and CH_4 (XCH_4) that can be monitored from space by dedicated sensors.

Limiting to XCO_2 estimation, which is the focus of this work, CarbonSpec [2] onboard of TANSAT and Orbiting Carbon Observatory-2 and -3 (OCO-2 [3] and OCO-3 [4]) are examples of satellite systems currently adopted. Both are grating spectrometers capable of measuring the radiance in the SWIR CO_2 absorption window with high spectral resolution (~ 0.1 nm) and spatial resolution of about 2 km.

The recent availability of satellite hyperspectral imaging sensor, such as PRISMA [5], [6] and EnMAP [7] operating in the visible near InfraRed (VNIR) and SWIR spectral range with a spectral resolution of 10 nm and a spatial resolution of 30 m, offers the opportunity to increase CO_2 monitoring capabilities. While they do not have the high spectral resolution of grating spectrometers and are not optimized for CO_2 mapping, they sample the CO_2 absorption window at 2000 nm with about 20 spectral channels and can potentially be used to quantify XCO_2 in the atmosphere at a finer spatial scale. This has motivated research activities aimed at analyzing the potential of the current generation of satellite hyperspectral imaging sensor for CO_2 retrieval [8], [9], [10] as well as this work where, with specific reference to the PRISMA sensor, a new algorithm for CO_2 estimation is proposed and a simulation-based sensitivity analysis is conducted to assess its performance.

A. Related Works

The methods adopted for CO_2 retrieval from satellite spectrometers (e.g., CarbonSpec and OCO-3) mainly exploit spectroscopy and can be classified as physics-based approaches. They rely on the physical relationships that model the radiative transfer in the atmosphere (radiative transfer model (RTM) [11]). RTM relates the at-sensor radiance to

Manuscript received 28 April 2023; revised 19 July 2023 and 11 September 2023; accepted 19 October 2023. Date of publication 26 October 2023; date of current version 7 November 2023. This work was supported by the Italian Space Agency (ASI) under Grant DC-UOT-2019-061. *(Corresponding author: Nicola Acito.)*

Nicola Acito, M. Alibani, and G. Corsini are with the Department of Information Engineering, University of Pisa, 56122 Pisa, Italy (e-mail: nicola.acito@unipi.it).

M. Diani is with the Accademia Navale di Leghorn—Italian Navy, 57127 Livorno, Italy.

Digital Object Identifier 10.1109/TGRS.2023.3327756

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the reflectance of the observed surface through radiative quantities that account for the parameters (e.g., XCO_2 and XCH_4) pertaining to the atmospheric constituents. Algorithms based on the RTM process the measured radiance in order to solve an inverse problem whose core is the estimation of the parameters that characterize the atmosphere at the time of the acquisition. One of the most effective physics-based techniques is the iterative maximum a posteriori differential optical absorption spectroscopy (IMAP-DOAS), which was proposed in [12] to jointly estimate the atmospheric content of different absorbers such as water vapor (H_2O), CH_4, and CO_2. IMAP-DOAS is an iterative algorithm that basically has the form of a Gauss–Newton optimization problem and combines an accurate physics model of the atmospheric absorption due to the constituents of interest, a polynomial model for the subtended reflectance, and a statistical characterization for the noise and the model inaccuracy. Although it was designed for the grating spectrometer SCIAMACHY [12] that samples the incoming radiation in the spectral window of interest with high spectral resolution, IMAP-DOAS formulation has been extended to imaging hyperspectral sensors with fewer bands (i.e., coarser spectral resolution) and finer spatial resolution. In [13], a slightly modified version of the algorithm was proposed for CH_4 concentration retrieval from data acquired by the airborne visible/InfraRed imaging spectrometer (AVIRIS). In a more recent work [14], IMAP-DOAS was extended for the retrieval of multiple gas species, including CH_4, CO_2, and H_2O, and applied to data acquired by the AVIRIS-next generation sensor (AVIRIS-NG). Cusworth et al. [8] presented the results obtained by applying IMAP-DOAS for quantifying power plant CO_2 emission from data acquired by both airborne [AVIRIS-NG and global airborne observatory (GAO)] and spaceborne hyperspectral data (PRISMA). IMAP-DOAS performance is in general quite sensitive to the spectral reflectance of the surface observed [13]. Furthermore, as stated by several authors [15], [16], the iterative form of the algorithm makes it computationally expensive, thus preventing real-time mapping and application to large scenes such as those typically acquired by satellite sensors. For this reason, IMAP-DOAS is generally used for estimating CH_4 and/or CO_2 concentrations in small areas surrounding known point source emitters [17].

A physics-based algorithm not suitable for data acquired by grating spectrometers but quite effective when applied to data acquired by hyperspectral sensors [9], [10] is the continuum interpolated band ratio (CIBR). It is based on the correlation between the values of the parameter characteristic of a given atmospheric absorber and the depth of the absorption feature due to that constituent. CIBR is computationally efficient, but its estimation performance is quite sensitive to various noise sources typically encountered in real data.

To break free from the complexity of the physics model and achieve higher computational efficiency, data-driven methods have been developed for application to hyperspectral data. Most of them are matched filter (MF)-based algorithms [15], [16], [17], [18], [19], [20] that can quantify greenhouse gas enhancement over large scenes in a computationally efficient way. Although MF-based algorithms have been applied mainly for the characterization of CH_4 point sources [16], [18], [19], [20], they also apply to the case of CO_2 emitters.

TABLE I
PRISMA SENSORS SPECIFICATIONS

Swath/FOV	30 km / 2.77°
Ground Sampling Distance	30 m
Spectral Range	VNIR: 400 – 1010 nm SWIR: 920 – 2505 nm
Spectral Width (FWHM)	≤ 12 nm
Radiometric Quantization	12 bit
VNIR SNR	> 200:1
SWIR SNR	> 100:1
MTF@Nyquist Freq.	VNIR/SWIR along Track > 0.18 VNIR/SWIR across Track > 0.34
Spectral Bands	66 VNIR 173 SWIR

MF-based algorithms linearize the RTM so as to view the at-sensor spectral radiance as the sum of two vectors: the first represents the background ($\boldsymbol{L}_0$), while the second is the contribution due to gas enhancement that is modeled as $\alpha \boldsymbol{t}$, where $\boldsymbol{t}$ is the target spectrum and α is the gas concentration enhancement over the background. $\boldsymbol{L}_0$ is assumed as a Gaussian random vector whose mean and covariance matrix are estimated from the image pixels (or in a neighborhood of the test pixel). $\boldsymbol{t}$ is obtained as the product of the background mean and a known unit enhancement vector denoting the change in transmittance corresponding to a change in concentration of the gas of interest. Thus, the problem reduces to the estimation of α that is obtained by projecting the data on the subspace addressed by $\boldsymbol{t}$ (after whitening by the background covariance matrix).

In contrast to physics-based techniques, MF-based algorithms are specifically conceived to estimate the emission enhancement (with respect to the background) from point sources and are not suitable to obtain a pixel-by-pixel estimation of the gas concentration.

In this work, we propose a new algorithm that estimates XCO_2 from satellite hyperspectral data on a per-pixel basis. It is a physics-based approach that, as opposed to IMAP-DOAS, separates the estimation of the unknowns so as to avoid the need for iterative processing. The proposed solution recasts the XCO_2 estimation problem in a linear form that allows the use of the MF-based approach. Thus, it exploits the strengths of both previously summarized approaches by allowing a per-pixel estimation of XCO_2 with a low computational burden. The algorithm is derived with specific reference to the PRISMA hyperspectral sensor (whose specifications are shown in Table I). However, it can be generalized to any satellite hyperspectral sensors operating in the VNIR–SWIR spectral range. To discuss the performance of the proposed technique, we perform an extensive analysis on a large dataset of simulated images obtained considering the specifications of the PRISMA sensor.

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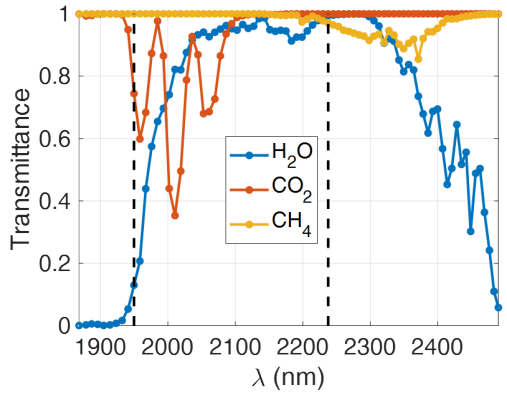


Fig. 1. Spectral transmittance due to H_2O, CO_2, and CH_4 as a function of the central wavelengths of PRISMA channels in the SWIR spectral range (channels from 150 to 230).

The rest of this article is organized as follows. In Section II, we describe the algorithm and discuss the assumptions made in its derivation. Then, in Section III, we provide a description of the simulation strategy for the generation of the image dataset adopted to analyze the performance of the algorithm. The results of such an analysis are provided in Section IV. An example on real data is presented in Section V. Finally, we draw the main conclusions of the work and underline the open issues by tracing the route for future developments.

II. Algorithm Description

This section introduces the RTM and the main hypotheses made and provides the detailed description of the proposed algorithm. Before delving into the mathematical formulation of the problem, let us look at Fig. 1, which provides an overview of the atmospheric absorption phenomena in the SWIR spectral range. Fig. 1 shows, in separated plots, the contributions of H_2O, CH_4, and CO_2 to the atmospheric transmittance. These three contributions, which act together on the incoming radiance, are shown with reference to the PRISMA channels from 150 to 230 (i.e., they were convolved with the spectral response functions (SRFs) of the considered channels of PRISMA and sampled at the nominal central wavelengths). Fig. 1 shows that CH_4 and CO_2 absorb the radiation into nonoverlapping spectral windows. Conversely, H_2O interacts with both CH_4 and CO_2 making it more difficult to estimate their amount in the atmosphere. In the following, the estimation of CO_2 is approached by considering the spectral channels from 159 (1950 nm) to 194 (2237 nm) so as to minimize the influence of CH_4. The spectral range to which we refer is highlighted in Fig. 1 by the two dashed vertical lines. Fig. 1 also provides evidence that, as stated in Section I, the number of PRISMA samples falling within the range of interest is much lower than the hundreds of spectral samples measured by high-resolution spectrometers (e.g., OCO-3 and CarbonSpec) specifically conceived for CO_2 monitoring.

A. RTM

According to the RTM in [11] and [21] and assuming the observed surface as Lambertian, the at-sensor radiance vector $\mathbf{L}\epsilon R^{N_B}$ for each pixel and for the N_B sensor channels of interest is expressed as follows:

$$\mathbf{L} = \boldsymbol{A}(\boldsymbol{\Gamma}) \cdot \boldsymbol{\rho} + \mathbf{L}_p(\boldsymbol{\Gamma}) + \boldsymbol{N}$$
$$\boldsymbol{A}(\boldsymbol{\Gamma}) = \frac{\mathbf{E}\cos(\vartheta)}{\pi} \cdot \boldsymbol{\tau}\ (\boldsymbol{\Gamma}) \quad (1)$$

where the symbol "$\cdot$" indicates the element-wise product. Equation (1) relates $\mathbf{L}$ to the surface spectral reflectance $\boldsymbol{\rho}$ and the spectral radiometric quantities $\mathbf{E}$, $\boldsymbol{\tau}$, and $\mathbf{L}_p$ that consider the effects of the atmosphere and solar irradiance. More precisely, $\mathbf{E}$ is the solar spectral irradiance at the top of the atmosphere, ϑ is the solar zenith angle, $\boldsymbol{\tau}$ considers the direct and diffuse solar radiation in both the downward and upward path, and $\mathbf{L}_p$ is the intrinsic upwelling atmospheric spectral radiance. $\boldsymbol{N}$ is the noise modeled as a zero-mean random vector.

It is important to note that the model in (1) (unlike [11] and [21]) does not include the contributions due to the background (adjacency effects) and the spectral spherical albedo because they are practically negligible in the spectral range of interest.

The radiometric quantities can be obtained by dedicated radiative transfer codes (RTC) as a function of a set of parameters [$\boldsymbol{\Gamma}$ in (1)] characterizing the acquisition geometry and the atmosphere. Throughout this article, we will refer to Modtran (version 5. [22]), which is one of the most popular commercial RTCs.

$\boldsymbol{\Gamma}$ can be divided into three subsets:

1) $\boldsymbol{\Gamma}_g$ that comprises all the acquisition geometry parameters such as the solar zenith (ϑ) angle, the sensor viewing angle, and the terrain elevation and slope;
2) $\boldsymbol{\Gamma}_s$ that includes the parameters that define the scattering effects such as the aerosol model and the visibility range (V); and
3) $\boldsymbol{\Gamma}_a$ that refers to the atmospheric absorbers, which, in the considered case, are H_2O quantified in terms of column H_2O concentration (w) usually measured in g/cm^2 and CO_2 quantified in terms of XCO_2 (hereinafter denoted as x) generally measured in parts per million (ppm).

In practice, the parameters $\boldsymbol{\Gamma}_g$, except for the terrain elevation and slope, are known as they can be obtained from the onboard instrumentation. The terrain elevation and slope are known if a digital elevation model (DEM) is available. When a DEM is not provided (as in the case of PRISMA data), the monitored surface is generally considered flat and at sea level. The scattering effects parametrized by $\boldsymbol{\Gamma}_s$ are featureless in the spectral domain (i.e., have a low-pass behavior) and have a more significant impact at shorter wavelengths. Therefore, SWIR channels are less affected by scattering. This motivated the simplifying assumption made in this work that the variability of scattering effects in the considered spectral range is quite small so as to set standard values for the parameters $\boldsymbol{\Gamma}_s$. Specifically, with reference to the parameters adopted by Modtran, we consider the rural aerosol model and the standard value $\bar{V} = 23$km for the visibility range.

The column H_2O concentration w generally varies within the monitored scene. A pixel-by-pixel estimation $\widehat{w}$ of w can be obtained by algorithms [22], [23], [24], [25] acting on the spectral windows where absorption effects are exclusively

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caused by H_2O (e.g., around 940 nm). In the rest of this article, we will refer to the well-known atmospheric precorrected differential absorption (APDA) algorithm proposed in [22].

B. RTM-Based MF Algorithm (RTM-MF) for XCO_2 Estimation

Considering the previous observations, the RTM in (1) is rewritten as follows:

$$\mathbf{L} = \boldsymbol{A}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x) \cdot \boldsymbol{\rho} + \mathbf{L}_{p,\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x) + \boldsymbol{N} \tag{2}$$

where $\bar{\boldsymbol{\Gamma}}$ groups the known parameters; the subscript indicates that the radiometric quantities refer to $\bar{\boldsymbol{\Gamma}}$ and their dependence on x and $\widehat{w}$ is made explicit.

According to the Beer–Lambert absorption law, $\boldsymbol{A}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x)$ can be expressed as follows:

$$\boldsymbol{A}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x) = \boldsymbol{\alpha}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}) \cdot e^{-\boldsymbol{\beta}_{\bar{\boldsymbol{\Gamma}}} x} \tag{3}$$

where $\boldsymbol{\alpha}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w})$ addresses the contributions of the solar radiation and the atmosphere in the absence of CO_2 and $\boldsymbol{\beta}_{\bar{\boldsymbol{\Gamma}}}$ considers the CO_2 absorbance and the path length. Using a first-order Taylor series expansion around a given value x_0, (3) can be linearized as follows:

$$\boldsymbol{A}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x) \cong \boldsymbol{A}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) - \boldsymbol{A}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) \cdot \boldsymbol{\beta}_{\bar{\boldsymbol{\Gamma}}} \cdot \Delta x \tag{4}$$

with $\Delta x = x - x_0$. Without loss of generality, we could consider for x_0 the current global value of XCO_2, i.e., $x_0 = 415$ppm [26].

In the SWIR spectral range, $\mathbf{L}_{p,\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x)$ is much lower than $\boldsymbol{A}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x)$ and it is neglected in most practical applications. Here, we do not ignore the contribution of the upwelling radiance, but we assume that its variation is negligible around x_0 so as to set $\mathbf{L}_{p,\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x) = \mathbf{L}_{p,\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0)$.

Consequently, and assuming for the moment that $\boldsymbol{\rho}$ is known, (2) is rewritten as follows:

$$\begin{aligned} \mathbf{L} &= \boldsymbol{c}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) + \boldsymbol{t}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) \cdot \Delta x + \mathbf{L}_{p,\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) + \boldsymbol{N} \\ &\text{with} \\ \boldsymbol{c}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) &= \boldsymbol{A}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) \cdot \boldsymbol{\rho} \\ \boldsymbol{t}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) &= -\mathbf{c}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) \cdot \boldsymbol{\beta}_{\bar{\boldsymbol{\Gamma}}} \end{aligned} \tag{5}$$

that reformulates the approached problem as a linear estimation problem whose minimum mean square error (MMSE) solution is given as follows:

$$\begin{aligned} \widehat{\Delta x} &= \frac{\boldsymbol{t}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0)^T \boldsymbol{Y}(\widehat{w}, x_0)}{\|\boldsymbol{t}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0)\|_2^2} \\ &\text{with} \\ \boldsymbol{Y}(\widehat{w}, x_0) &= \mathbf{L} - \boldsymbol{c}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) - \mathbf{L}_{p,\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0) \end{aligned} \tag{6}$$

where ^{T} is the vector transpose operator and $\|\|_2$ is the l_2-norm. Equation (6) can be interpreted as the application of the MF $(\boldsymbol{t}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0))/(\|\boldsymbol{t}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0)\|_2^2)$ to the observation vector $\boldsymbol{Y}(\widehat{w}, x_0)$.

Of course, the estimate $\widehat{x}$ of XCO_2 is obtained as follows:

$$\widehat{x} = x_0 + \widehat{\Delta x}. \tag{7}$$

It is worth noting that (6) is applied on a pixel-by-pixel basis. However, when we can assume that the parameter x is constant in an MxM pixels neighborhood Ω of the pixel under test, a more robust estimator can be obtained as follows:

$$\widehat{\Delta x} = \frac{\sum_{k\in\Omega} \boldsymbol{t}_{\bar{\boldsymbol{\Gamma}}}^{(k)}\left(\widehat{w}^{(k)}, x_0\right)^T \boldsymbol{Y}^{(k)}\left(\widehat{w}^{(k)}, x_0\right)}{\sum_{k\in\Omega} \left\|\boldsymbol{t}_{\bar{\boldsymbol{\Gamma}}}^{(k)}\left(\widehat{w}^{(k)}, x_0\right)\right\|_2^2} \tag{8}$$

where $\boldsymbol{Y}^{(k)}$, $\boldsymbol{t}_{\bar{\boldsymbol{\Gamma}}}^{(k)}$, and $\widehat{w}^{(k)}$ are the observation vector, the MF, and the estimate of w for the kth pixel of Ω, respectively.

Up to now, to derive the MF-based XCO_2 estimation algorithm, we have assumed that $\boldsymbol{\rho}$ is known. Of course, such an assumption is not realistic and a possible way to deal with this problem is to estimate the surface reflectance from the data themselves. For this purpose, we exploit the idea proposed in two recently published papers [21], [27] where machine learning methods are used to directly estimate the spectral reflectance from the at-sensor radiance addressing the variability due to one or more parameters of the RTM (and the noise) without estimating them.

Following the mentioned approach, we define a parametric transformation that allows us to obtain the estimate $\widehat{\boldsymbol{\rho}}$ of the spectral reflectance in each pixel from the observation $\tilde{\boldsymbol{\rho}}(x)$,which is obtained by inverting the model in (2) with a fixed value of XCO_2 (e.g., x_0). To be more precise, let us start by defining the observation vector $\tilde{\boldsymbol{\rho}}(x)$ as follows:

$$\tilde{\boldsymbol{\rho}}(x) = \frac{\mathbf{L} - \mathbf{L}_{p,\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0)}{\boldsymbol{A}_{\bar{\boldsymbol{\Gamma}}}(\widehat{w}, x_0)}. \tag{9}$$

It can be seen as a rough estimate of the spectral reflectance, which contains the effects of noise and does not consider the actual value of XCO_2 (this is made explicit in (9) through the dependence on x).

$\widehat{\boldsymbol{\rho}}$ is obtained by applying the parametric estimation function g to $\tilde{\boldsymbol{\rho}}(x)$, i.e.,

$$\widehat{\boldsymbol{\rho}} = g\left(\tilde{\boldsymbol{\rho}}(x); \boldsymbol{\Psi}\right) \tag{10}$$

with $\boldsymbol{\Psi}$ denoting the parameters of the estimation function. The response of the estimator to the predictors $\tilde{\boldsymbol{\rho}}(x)$ depends on the specific value of the parameters $\boldsymbol{\Psi}$ that must be determined through the learning process. Learning is performed on N_t training samples, made up of the predictors $\tilde{\boldsymbol{\rho}}_{\boldsymbol{t}}(x)$ and the corresponding reflectance vector $\boldsymbol{\rho}_t$ $(t = 1, \ldots, N_t)$. They are obtained by simulation using the RTM in (2), Modtran for the generation of the radiometric quantities and by exploiting the availability of a database of the spectral reflectances of different surfaces/materials. The learning process consists in minimizing with respect to $\boldsymbol{\Psi}$ the loss function in (11) over the training samples

$$\begin{aligned} &\text{loss}\left[g\left(\tilde{\boldsymbol{\rho}}_{\boldsymbol{t}}(x); \boldsymbol{\Psi}\right), \boldsymbol{\rho}_t\right] \\ &\qquad t = 1, \ldots, N_t \\ &= \sum_{t=1}^{N_t} \left\|\boldsymbol{\rho}_t - g\left(\tilde{\boldsymbol{\rho}}_{\boldsymbol{t}}(x); \boldsymbol{\Psi}\right)\right\|_2^2 + \phi\|\boldsymbol{\Psi}\|_F^2. \end{aligned} \tag{11}$$

The loss function is composed of a data fidelity term that computes the square estimation error and a regularization term based on the Frobenious norm $(\|\|_F)$ of $\boldsymbol{\Psi}$, which is introduced to avoid overfitting and increase robustness to the

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noise. According to the results in [21] and [27], we consider a multilinear parametric estimator

$$\widehat{\boldsymbol{\rho}} = \boldsymbol{\Psi}\begin{bmatrix}\tilde{\boldsymbol{\rho}}(x)\\ 1\end{bmatrix} \tag{12}$$

which gives the loss function in (12) the form of a Ridge regression whose closed-form solution is

$$\boldsymbol{\Psi} = \left[\sum_{t=1}^{N_t}\begin{bmatrix}\tilde{\boldsymbol{\rho}}_t(x)\\ 1\end{bmatrix}\begin{bmatrix}\tilde{\boldsymbol{\rho}}_t(x)\\ 1\end{bmatrix}^T + \phi\cdot\mathbf{I}\right]^{-1}\cdot\sum_{t=1}^{N_t}\begin{bmatrix}\tilde{\boldsymbol{\rho}}_t(x)\\ 1\end{bmatrix}\boldsymbol{\rho}_t^T \tag{13}$$

with $\mathbf{I}$ denoting the (N_B+1)x(N_B+1) identity matrix.

Details on training set generation are provided in Section III, and here, we would like to conclude with a summary of the RTM-based MF algorithm (RTM-MF, hereinafter) proposed in this section. First, for each image pixel, RTM-MF derives the estimation of $\widehat{w}$ by applying the APDA algorithm to the channels falling in the H_2O absorption window around 940 nm. Then, it estimates the spectral reflectance $(\widehat{\boldsymbol{\rho}})$ by applying (12) to the observation vector $\tilde{\boldsymbol{\rho}}(x)$. The latter is obtained from the at-sensor radiance in the selected CO_2 absorption bands (Fig. 1) by means of (9). Finally, the MF-based estimator in (6) [or that in (8)] is applied to each image pixel (or to its surrounding) in order to obtain the estimate $\widehat{x}$ of XCO_2.

III. Simulation Strategy

This section describes the simulation strategy for the generation of the PRISMA images (PRISMA-like images, hereinafter) adopted for the performance analysis and presents the procedure followed to build the training set used in the learning process of the reflectance estimator introduced in Section II.

A. Simulation of PRISMA-Like Images

Note that our simulator generates synthetic images covering the whole spectral range of the PRISMA sensor. However, only a small portion of the SWIR and NIR spectral regions is actually used. In fact, the RTM-MF procedure leverages the spectral channels falling between 1950 and 2237 nm, while to estimate w, we exploit a small spectral region around 940 nm.

The simulation of each PRISMA-like image starts from a real reflectance image that is picked up from an AVIRIS-NG image repository made available by the NASA Jet Propulsion Laboratory [28]. AVIRIS-NG sensor covers the VNIR–SWIR spectral range from 380 to 2510 nm (with 5-nm spectral sampling) that completely overlaps the portion of the spectrum measured by PRISMA. In this work, we use ten AVIRIS-NG reflectance images acquired during the India Campaign held in 2016. From each image, we extracted nonoverlapped tiles of 512 × 512 pixels (100 in total), discarding cloud-covered areas and corrupted data. A detailed list of the characteristics (including the file identifiers and the number of tiles extracted) of the ten AVIRIS-NG images is given in Appendix I, where the false-color (FC) representation of one of the images is also shown along with the extracted tiles. The reflectance spectra in each image were low-pass filtered in order to remove residual noise and interpolated so as to achieve a 1-nm spectral sampling.

For a given reflectance image, the atmospheric effects are simulated to obtain the at-sensor radiance. This task is performed by applying the RTM in [21], with the radiometric quantities obtained by specific lookup tables (LUTs) generated by Modtran (with 1-nm spectral sampling) for various combinations of the significant parameters. In all the simulations, the "Mid-Latitude Summer" model of Modtran is adopted. Table II lists the parameters related to the acquisition geometry and the scattering effects and the corresponding values considered in the simulations.

As to the absorption effects, both w and x are generated in terms of bidimensional (2)-D) maps with the same spatial extent as the image so as to consider their spatial variability within the scene. Specifically, with reference to w, first, we generate a 2-D map whose pixels are realizations of independent random variables uniformly distributed in the range $[0, 1]$. Then, we introduce a spatial correlation by applying a 2-D linear (zero phase) space-invariant filter. The amplitude response of such a filter is defined according to the average power spectrum of the maps obtained by applying APDA to dozens of real PRISMA images. Finally, the generated spatially correlated map is rescaled in the range $[w_{LB}, w_{UB}]$ with w_{LB} and w_{UB} in turn randomly selected in the range $[0.1, 5]^{g}/_{cm^2}$.

A similar procedure is adopted to generate the maps of XCO_2. In this case, the spatial correlation is obtained by applying the same linear filter of the previous case followed by a 2-D moving window average filter so as to simulate the higher spatial correlation expected in the x maps. The obtained correlated maps are rescaled in the range $[x_{LB}, x_{UB}]$, with x_{LB} and x_{UB} randomly selected in the range [360, 550]ppm.

Fig. 2(a) and (b) shows an example of the generated maps for the parameters w and x, respectively.

Once the high spectral resolution radiance image is obtained, the at-sensor radiance measured by PRISMA is derived by applying the SRFs of each channel and adding noise. The SRF of each channel is modeled as a Gaussian kernel centered at the nominal central wavelength and having width in agreement with the nominal full-width half-maximum (FWHM) of that channel. Both the mentioned parameters are known as they are included in the metadata provided with the PRISMA images.

Noise is generated according to the model in [29] as a spectrally uncorrelated Gaussian random vector with zero mean and variance per band linearly dependent on the noise-free radiance in order to consider both the photon noise and the thermal noise contributions. Specifically, for the bth channel of a given pixel, noise is obtained as a realization of a zero-mean Gaussian variable with variance

$$\sigma_b^2 = \gamma_{ph,b}L_b + \gamma_{th,b}. \tag{14}$$

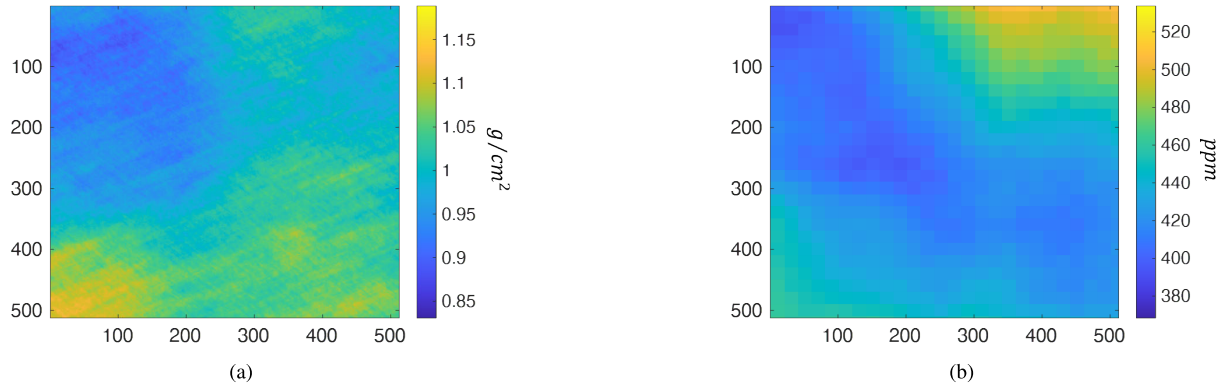
In (14), L_b is the noise-free radiance in the bth channels, while $\gamma_{ph,b}$ and $\gamma_{th,b}$ are the parameters related to photon and thermal noise, respectively. To make noise simulation realistic,

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TABLE II
LIST OF THE PARAMETERS RELATED TO THE ACQUISITION GEOMETRY AND THE SCATTERING EFFECTS CONSIDERED IN THE SIMULATION

Acquisition Geometry	
Parameter	**Values** (lower bound : step size : upper bound)
Solar Zenith Angle (deg)	0 : 10 : 40
Sensor Viewing Angle (deg)	0 : 5 : 35
Terrain elevation (m)	0

Scattering	
Parameter	**Values**
Aerosol model (IHAZE in Modtran)	1 (Rural), 3 (Navy Maritime), 5 (Urban)
Visibility (km)	10:5:60 (lower bound : step size : upper bound)



(a) (b)

Fig. 2. Examples of simulated (a) w and (b) x maps.

for both $\gamma_{ph,b}$ and $\gamma_{th,b}$, we use the average values estimated from dozens of real PRISMA image by the algorithm proposed in [29].

The simulation strategy has been used to simulate $N_{IM} = 840$ PRISMA-like images by selecting randomly the reflectance images, the combinations of the parameters in Table II, and by generating the maps of w and x according to the previously described procedure. As an example, Fig. 3 shows the FC representation, which is obtained by combining the channels corresponding to red, green, and blue, of one of the generated images.

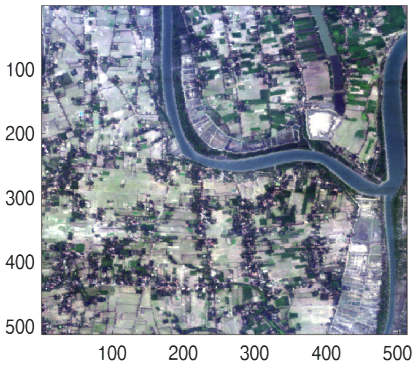


Fig. 3. FC representation of one of the simulated PRISMA-like images.

B. Simulation of the Training Set for the Reflectance Estimator

Regarding the learning of the reflectance estimator, the training set is obtained by a procedure similar to that previously described, except for the generation of the surface reflectances. The tth reflectance spectrum $\boldsymbol{\rho}_t$ is generated by exploiting a library $\mathbf{D}$, which contains more than 1000 measured spectral reflectances of several man-made materials, vegetation, water, soils, and minerals. $\mathbf{D}$ is obtained by merging the three available spectral libraries ASTER [30], USGS (version released in 2007, [31]), and ANGERS [32].

Each $\boldsymbol{\rho}_t$ is obtained by combining N randomly selected elements of $\mathbf{D}$ ($\mathbf{d}_i, i = 1, \ldots, N$) and N randomly generated mixing coefficients a_i, $i = 1, \ldots, N$

$$\boldsymbol{\rho}_t = \sum_{i=1}^{N} a_i \mathbf{d}_i. \tag{15}$$

The a_is are drawn according to the Dirichlet distribution so as to meet the sum-to-one and the nonnegative constraints. N is also generated as a random integer in the range [1], [5]. This strategy increases the spectral variability of the training set with the aim of encompassing the wide range of spectral reflctances encountered in real applications.

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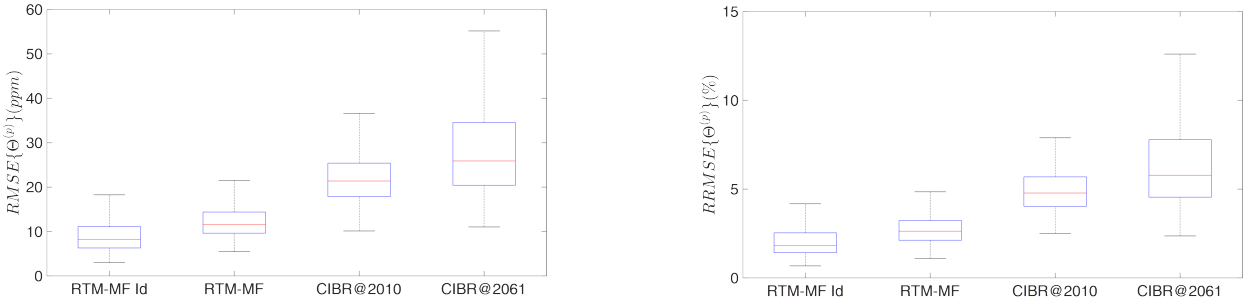


Fig. 4. Distribution in terms of boxplot of (a) RMSE$\left\{\Theta^{(p)}\right\}$ expressed in ppm and (b) RRMSE$\left\{\Theta^{(p)}\right\}$ in percent for the algorithms: RTM-MF Id, RTM-MF, CIBR@2010, and CIBR@2061.

For each $\boldsymbol{\rho}_t$, the corresponding radiance is obtained according to the above-described simulation strategy, with w and x randomly drawn in the range $[0.1, 5]^{g}/_{cm^2}$ and [330, 550]ppm, respectively. Random noise is added to the simulated radiance according to the model in (14).

Of course, the obtained radiance is processed according to (9) to obtain the tth training pair $\left\{\boldsymbol{\rho}_t, \tilde{\boldsymbol{\rho}}_t(x)\right\}$.

By means of the described procedure, we simulated $N_t = 5 \times 10^4$ training samples. It is worth noting that the simulation of the training set (in particular, that of the reflectance data) is made different from that of the PRISMA-like images in order to have training and test sets as independent as possible.

IV. Results on PRISMA-Like Data

This section is dedicated to the analysis of the performance of the proposed RTM-MF algorithm. This analysis is carried out on the test set consisting of $N_{IM} = 840$ PRISMA-like images generated by the simulation strategy described in Section III.

For completeness, RTM-MF is compared with the CIBR algorithm that provides pixel-by-pixel estimates of XCO_2 by exploiting the correlation between XCO_2 values and the depth of the absorption feature in a specific measurement channel. More precisely, for a given measurement channel m within the CO_2 absorption window, CIBR computes the depth of the absorption feature relative to a local linear continuum as follows:

$$\tilde{d} = \frac{L_m}{w_1 L_{r1} + w_2 L_{r2}}. \tag{16}$$

$r1$ and $r2$ are two channels at the left and right sides of the absorption window where the effect of CO_2 is negligible. The two weighting coefficients w_1 and w_2 are nonnegative and sum to unity so as to make the denominator the linearly interpolated continuum at the location m of the absorption window.

According to the RTM in (2), neglecting the upwelling radiance and assuming as constant the reflectance values in the three channels m, $r1$, and $r2$, the theoretical values of the absorption window depth as a function of x are expressed as follows:

$$d(x) = \frac{A_{\bar{\boldsymbol{\Gamma}},m}(\widehat{w}, x)}{w_1 A_{\bar{\boldsymbol{\Gamma}},r1}(\widehat{w}) + w_2 A_{\bar{\boldsymbol{\Gamma}},r1}(\widehat{w})}. \tag{17}$$

$d(x)$ can be computed by Modtran for different values of x so as to estimate the XCO_2 value in each pixel as follows:

$$\widehat{x} = \arg\min_x \left|\tilde{d} - d(x)\right|. \tag{18}$$

In the tested implementation of the CIBR algorithm, we consider $r1 = 163$ and $r2 = 178$ corresponding to the PRISMA channels at 1985 and 2110 nm, respectively. As to the measurement channel, we tested two different choices: 1) $m = 166$ (2010 nm) as suggested in [10] and 2) $m = 172$ (2061 nm) as indicated in [9]. In the following, we will refer to the two implementations as CIBR@2010 and CIBR@2061.

It is worth noting that, as in the case of RTM-MF, when we can assume x constant in a neighborhood Ω of the pixel under test, a more robust CIBR estimator can be obtained as follows:

$$\tilde{d} = \frac{\sum_{k\in\Omega} L_m^{(k)}}{w_1 \sum_{k\in\Omega} L_{r1}^{(k)} + w_2 \sum_{k\in\Omega} L_{r2}^{(k)}} \tag{19}$$

where $L_m^{(k)}$, $L_{r1}^{(k)}$, and $L_{r2}^{(k)}$ are radiance values in the kth pixel of Ω.

In all the experiments presented in this section, for RTM-MF and the two CIBR algorithms, we consider the robust versions with Ω defined as a 3×3 pixels neighborhood. In applying the RTM-MF algorithm, the pivot point x_0 of the first-order Taylor approximation of $e^{-\boldsymbol{\beta}_{\bar{\Gamma}} x}$ [see (3) and (4)] is set to 415 ppm according to the current global mean value of XCO_2 [26]. Furthermore, to evaluate the impact of the estimation $\widehat{\boldsymbol{\rho}}$ (10) on the performance of the RTM-MF, we consider an ideal version of the algorithm (named "RTM-MF Id") in which the true value of $\boldsymbol{\rho}$ is used instead of $\widehat{\boldsymbol{\rho}}$.

In all the discussed results, we discarded the pixels with reflectance lower than 0.03 because the variations of the corresponding radiance values with respect to x are less than the sensitivity of any algorithm considered.

In order to test the performance of the various XCO_2 estimation methods, we analyze the random variable $\Theta = \widehat{x} - x$, which corresponds to the estimation error.

The first analysis is focused on the root-mean-square error (RMSE) and the root relative mean square error (RRMSE) of Θ obtained on each test image. For the pth image, RMSE and

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RRMSE are defined as follows:

$$\mathrm{RMSE}\{\Theta^{(p)}\} = \sqrt{\frac{1}{Np}\sum_{k=1}^{Np}\left[\widehat{x}_k^{(p)} - x_k^{(p)}\right]^2}$$

$$\mathrm{RRMSE}\{\Theta^{(p)}\} = \sqrt{\frac{1}{Np}\sum_{k=1}^{Np}\left[\frac{\widehat{x}_k^{(p)} - x_k^{(p)}}{x_k^{(p)}}\right]^2} \quad (20)$$

where $x_k^{(p)}$ and $\widehat{x}_k^{(p)}$ are the true and the estimated values of XCO_2 in the kth pixel of the image, respectively, and Np is the number of image pixels considered.

Fig. 4(a) and (b) shows the distribution in terms of boxplot of RMSE [Fig. 4(a)] and RRMSE [Fig. 4(b)] all over the considered images for the four algorithms RTM-MF Id, RTM-MF, CIBR@2010, and CIBR@2061. Each box plot provides the following information about the distribution of the index to which it refers: the red line is the median, the edges of the box are the 25th and the 75th percentiles, respectively, and the whiskers are the minimum and the maximum values.

Fig. 4(a) shows that over the whole dataset, aside for the ideal case, the proposed RTM-MF achieves the best performance in terms of RMSE. On 75% of the images, the values of RMSE obtained by RTM-MF are lower than 14.3 ppm with x ranging from 360 to 550 ppm. Conversely, CIBR@2010 provides RMSE values higher than 17 ppm on 75% of the analyzed images. The median RMSE attained by RTM-MF is of about 11 ppm, while that of CIBR@2010 is approximately 21 ppm.

By comparing the distributions of the RMSE in the ideal case (RTM-MF Id) and in the more realistic case (RTM-MF), we can see that, as expected, RTM-MF Id performs better than RTM-MF. In the ideal case, the RMSE is below 11 ppm on 75% of the images and its median value is 8.17 ppm. In general, by using the estimated value of $\boldsymbol{\rho}$, RTM-MF has a performance loss of about 3.5 ppm on average.

Fig. 4(a) also shows that CIBR@2010 outperforms CIBR@2061. This can be explained by considering that the strength of the CO_2 absorption is greater at 2010 nm than 2061 nm. It is worth noting that the absorption feature at 2061 nm is less sensitive to H_2O than that at 2010 nm. Therefore, CIBR@2061 should be preferred to CIBR@2010 when H_2O cannot be estimated and compensated for. Since, in our experiments, the estimate (via APDA) of the H_2O is available, in the rest of this article, we only discuss the performance of CIBR@2010.

Fig. 4(b) confirms all the conclusions drawn. In particular, it shows that RTM-MF attains RRMSE values below 3% on most of the analyzed images with a median of about 2.6%. Conversely, RRMSE for CIBR@2010 is greater than 4% for most images. Fig. 4(b) also confirms the degradation of the performance of RTM-MF with respect to the ideal case. To give a numerical example, the median RRMSE for ideal case drops to 1.83%.

In deriving RTM-MF, we used the first-order Taylor approximation of $e^{-\boldsymbol{\beta}_{\mathrm{T}}x}$ in (4) around the pivot point x_0. To test the validity of this hypothesis, we analyze the estimation error Θ conditioned to XCO_2 ($\Theta|x$). In particular, Fig. 5(a) shows the RMSE of $\Theta|x$ ($\mathrm{RMSE}\{\Theta|x\}$) for all the ranges of x values considered in our simulation. The values of $\mathrm{RMSE}\{\Theta|x\}$ achieved by RTM-MF are compared with those obtained by CIBR@2010 and those achieved in the ideal case. As to RTM-MF, Fig. 5(a) shows that, in the ideal case (RTM-MF Id), $\mathrm{RMSE}\{\Theta|x\}$ is quite stable as all index values are very close to 10.2 ppm. This leads to conclude that the error due to the first-order Taylor approximation of the term $e^{-\boldsymbol{\beta}_{\mathrm{T}}x}$ in (4) does not significantly affects the performance of the algorithm. As to the realistic case (RTM-MF), we can see that, for most of the x values, $\mathrm{RMSE}\{\Theta|x\}$ is concentrated around 13.2 ppm. It deviates from 13.2 ppm when approaching the limits of the considered range, where, probably, the error introduced by the estimation of $\boldsymbol{\rho}$ has the greatest impact. However, for all x considered, RTM-MF provides values of $\mathrm{RMSE}\{\Theta|x\}$ well below those achieved by CIBR@2010.

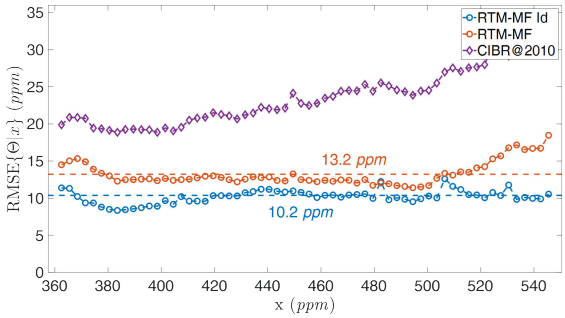
In addition to the RMSE analysis, Fig. 5(b) and (c) shows the results concerning the bias and the standard deviation of the estimators. In particular, Fig. 5(b) shows the values of the bias for each algorithm, which represent the average value of Θ conditioned to XCO_2 ($E\{\Theta|x\}$). On the other hand, Fig. 5(c) shows the standard deviation of $\Theta|x$ ($std\{\Theta|x\}$) for the considered algorithms. The bias exhibited by all the algorithms is not constant and varies with the specific value of x. CIBR@2010 has negative bias for each value of x with values ranging from -3 to -17.8 ppm, while the biases for RTM-MF Id and RTM-MF have both negative and positive values. Regardless of x, the absolute values of the bias obtained by CIBR@2010 are higher than those obtained by RTM-MF and RTM-MF Id. However, notice that the worse performance attained by CIBR@2010 in terms of RMSE is not solely due to the estimation bias. In fact, Fig. 5(c) shows that CIBR@2010 exhibits a significantly higher standard deviation of $\Theta|x$ compared to both RTM-MF and RTM-MF Id. As expected, Fig. 5(c) also shows that $std\{\Theta|x\}$ of RTM-MF Id is lower than that of RTM-MF.

The other hypothesis made in the algorithm derivation concerns the variability with respect to V of the radiometric quantities in the RTM. Specifically, it is assumed negligible and the standard value $\bar{V} = 23$km is used. In order to evaluate the impact of this hypothesis on the performance of RTM-MF, we analyze the RMSE of Θ conditioned to V ($\mathrm{RMSE}\{\Theta|V\}$) on the whole dataset. The results of such an analysis are summarized in Fig. 6, where, for completeness, $\mathrm{RMSE}\{\Theta|V\}$ obtained by CIBR@2010 and in the ideal case is also reported.

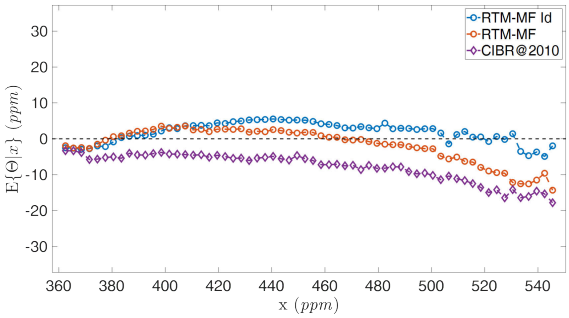
Fig. 6 shows that for V greater than 20km, $\mathrm{RMSE}\{\Theta|V\}$ for RTM-MF is quite constant (approximately 13.3 ppm), while it increases for lower values of V and rises up to 20 ppm with $V = 10$km. This leads us to conclude that the hypothesis made can be considered valid for almost all the visibility conditions that occur in real applications. It is less reasonable for very low visibilities. In those cases, the assumption made has a negative effect on the performance of the RTM-MF algorithm. It is worth noting that Fig. 6 confirms the better results achieved by RTM-MF compared to CIBR@2010 for almost all the values of V considered. When very low V values occur, RTM-MF and CIBR@2010 have very similar performance.

Finally, we evaluate the performance of the algorithms with respect to the average value $\bar{\boldsymbol{\rho}}$ of the reflectance in the spectral

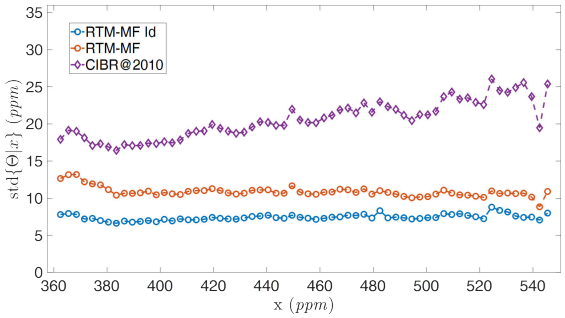
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(a)



(b)



(c)

Fig. 5. (a) RMSE of the estimation error conditioned to the values of x ($\Theta|x$) estimated on the entire dataset for RTM-MF Id, RTM-MF, and CIBR@2010. (b) Mean value of $\Theta|x$ and (c) standard deviation of $\Theta|x$ estimated on the entire dataset for RTM-MF Id, RTM-MF, and CIBR@2010.

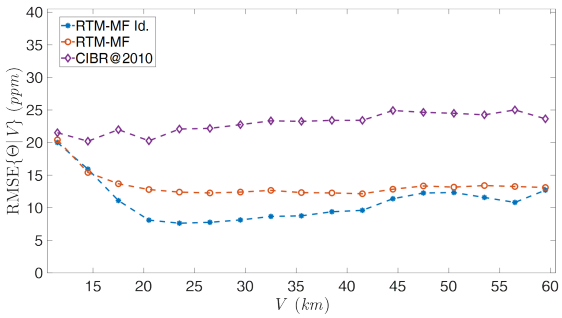


Fig. 6. RMSE of the estimation error conditioned to the values of V ($\Theta|V$) estimated on the entire dataset for RTM-MF Id, RTM-MF, and CIBR@2010.

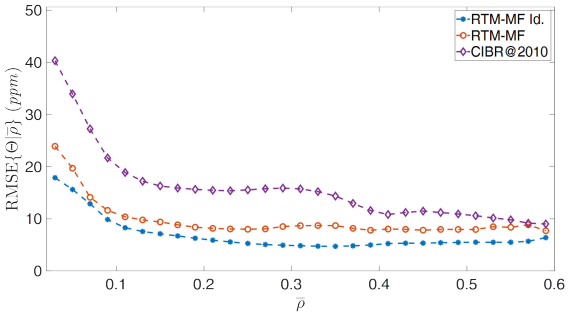


Fig. 7. RMSE of the estimation error conditioned to the values of the spectral reflectance average value ($\Theta|\bar{\rho}$) estimated on the entire dataset for RTM-MF Id, RTM-MF, and CIBR@2010.

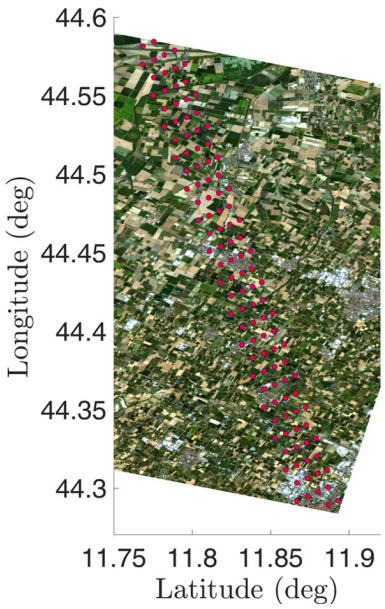
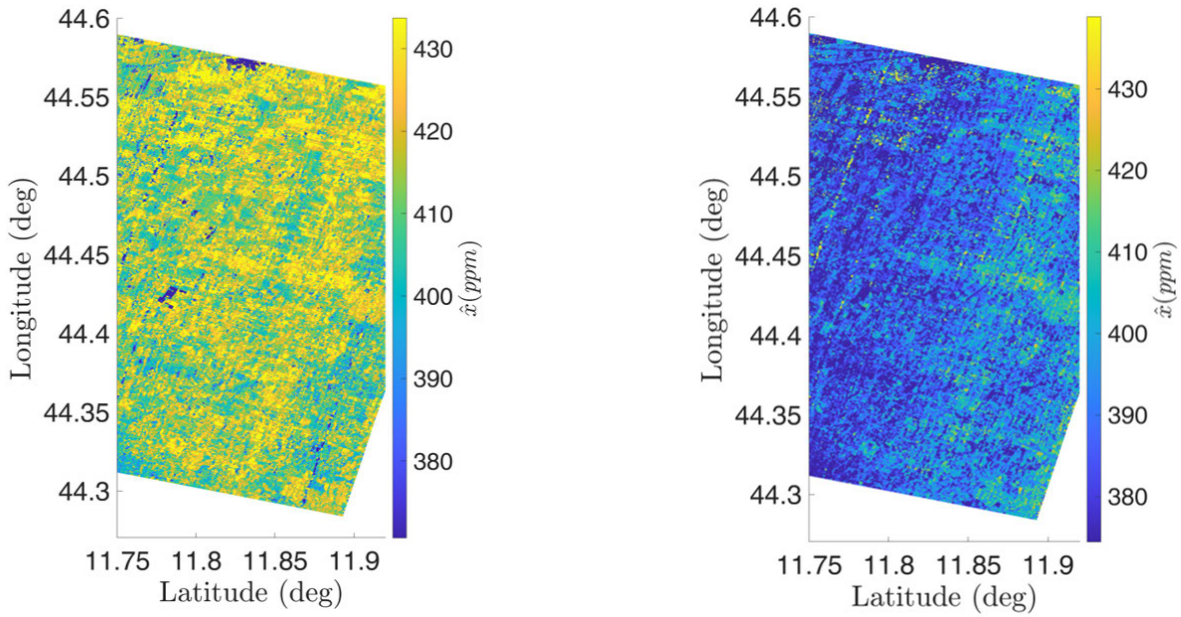


Fig. 8. False-color representation of the portion of the PRISMA image acquired on 15 July 2022 over the Po-Valley, Italy. Red circles indicate the coordinates of the OCO-2 data samples.

range of interest. Specifically, we analyze the RMSE of Θ conditioned to $\bar{\boldsymbol{\rho}}$ (RMSE$\{\Theta|\bar{\boldsymbol{\rho}}\}$). The results of such analysis are reported in Fig. 7. For all the algorithms, RMSE$\{\Theta|\bar{\boldsymbol{\rho}}\}$ decreases as $\bar{\boldsymbol{\rho}}$ increases. It is an expected behavior as the impact of the noise tends to decrease as $\bar{\boldsymbol{\rho}}$ increases. Fig. 7 shows that RTM-MF outperforms CIBR@2010 for all the values of $\bar{\boldsymbol{\rho}}$ occurred in the analyzed dataset. RMSE values for the two algorithms converge approximately to the same value for high $\bar{\boldsymbol{\rho}}$, while they differ significantly for lower values of $\bar{\boldsymbol{\rho}}$ (<0.4), thus showing the higher noise robustness of RTM-MF.

The performance of the algorithms degrades significantly with $\bar{\boldsymbol{\rho}} < 0.1$. Specifically, with $\bar{\boldsymbol{\rho}} < 0.1$, RTM-MF attains

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(a) (b)

Fig. 9. Estimates of XCO_2 obtained on the considered PRISMA image by (a) RTM-MF and (b) CIBR@2010.

RMSE$\{\Theta|\bar{\rho}\}$ higher than 11 ppm, and the value of the index rises up to 24 ppm for $\bar{\rho}$ in the order of 0.03 ppm.

We would like to conclude this section by providing information about the computational time of the proposed algorithm. By using a MATLAB implementation of the RTM-MF algorithm on a personal computer equipped with Intel Core i9 CPU at 2.5 GHz and with 64 GB of RAM, we found an average computing time of about 6s for a full image of 512×512 pixels. The MATLAB implementation of the CIBR algorithm takes about 4s on average.

V. Example on Real PRISMA Data

To give an example on real data, we considered a portion of the PRISMA image acquired on a subregion of the Po Valley-Italy on 15 July 2022. The FC representation of the considered image obtained by combining the channels corresponding to red, green, and blue is shown in Fig. 8. Such image has been chosen due to the availability of OCO-2 data acquired on the same day and in the same region. The longitude and latitude coordinates of OCO-2 data samples are highlighted in Fig. 8 (red plots).

Fig. 9(a) and (b) shows per-pixel estimates $\widehat{x}$ of XCO_2 obtained on the portion of the PRISMA image by RTM-MF and CIBR@2010, respectively. Both the algorithms are applied to PRISMA L1 Hyperspectral COregistered (HCO) data, i.e., the radiance data that are spatially and spectrally resampled in order to correct smile and keystone effects and spatial/spectral misalignments.

Fig. 9(a) and (b) shows that RTM-MF and CIBR@2010 provide different results. Specifically, the estimated values obtained by RTM-MF are in average higher than those provided by CIBR@2010.

Fig. 10 shows the scatterplot of the XCO_2 values $\bar{x}$ for each sample of OCO-2 data and the corresponding estimates

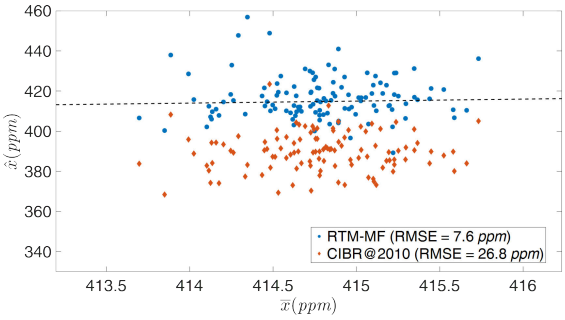


Fig. 10. Scatterplot of the XCO_2 values $\bar{x}$ for each sample of OCO-2 data and the corresponding estimates $\widehat{x}$ obtained by RTM-MF (blue points) and CIBR@2010 (red points).

$\widehat{x}$ obtained by RTM-MF (blue points) and CIBR@2010 (red points). To consider the different spatial resolutions of OCO-2 and PRISMA, the estimates $\widehat{x}$ shown in Fig. 10 are averaged over a 5×5 pixels neighborhood. Comparing XCO_2 estimates obtained by the two considered algorithms with those provided by OCO-2, it is evident that CIBR@2010 underestimates XCO_2 by providing an average value for the error $\widehat{x} - \bar{x}$ of about -25 ppm and RMSE of 26.8 ppm. Results improve by adopting RTM-MF that provides an average error of -1.9 ppm and RMSE of 7.6 ppm.

We would like to remark that the results presented in this section are just an example of the application of the proposed algorithm and do not have the claim to be an experimental proof of its effectiveness. There are some specific aspects related to real data that are not considered in this work and deserve further analysis. For instance, we have not yet investigated the impact of: 1) spectral calibration errors that generally affect real hyperspectral data; 2) instrumental fixed pattern noise (e.g., striping); and 3) not correctly compensated

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TABLE III

LIST OF AVIRIS-NG IMAGES ADOPTED FOR THE SIMULATED DATASET. FOR EACH IMAGE, THE TABLE GIVES: THE NAME OF THE FILE CORRESPONDING TO THE IMAGE (FILE IDENTIFIER), THE SENSING DATE, THE NOMINAL GSD (IN M), THE NUMBER OF LINES AND SAMPLES, AND THE NUMBER OF EXTRACTED TILES (512 × 512 PIXELS)

File Identifier	*Sensing date (day/month/year)*	*GSD (m)*	*Lines*	*Samples*	*No of extracted tiles*
ang20160101t072436	01/01/2016	5	7891	693	14
ang20160102t061817	02/01/2016	5	7836	694	14
ang20160102t064351	02/01/2016	5	7761	682	14
ang20160223t060914	23/02/2016	5	3591	711	4
ang20160223t062639	23/02/2016	5	3657	718	5
ang20160224t053740	24/02/2016	5	2670	671	4
ang20160224t055506	24/02/2016	5	4710	676	8
ang20160305t062710	05/03/2016	5	9177	756	12
ang20160305t063928	05/03/2016	5	9167	747	13
ang20160305t065158	05/03/2016	5	9086	768	12
Total no of extracted tiles					**100**

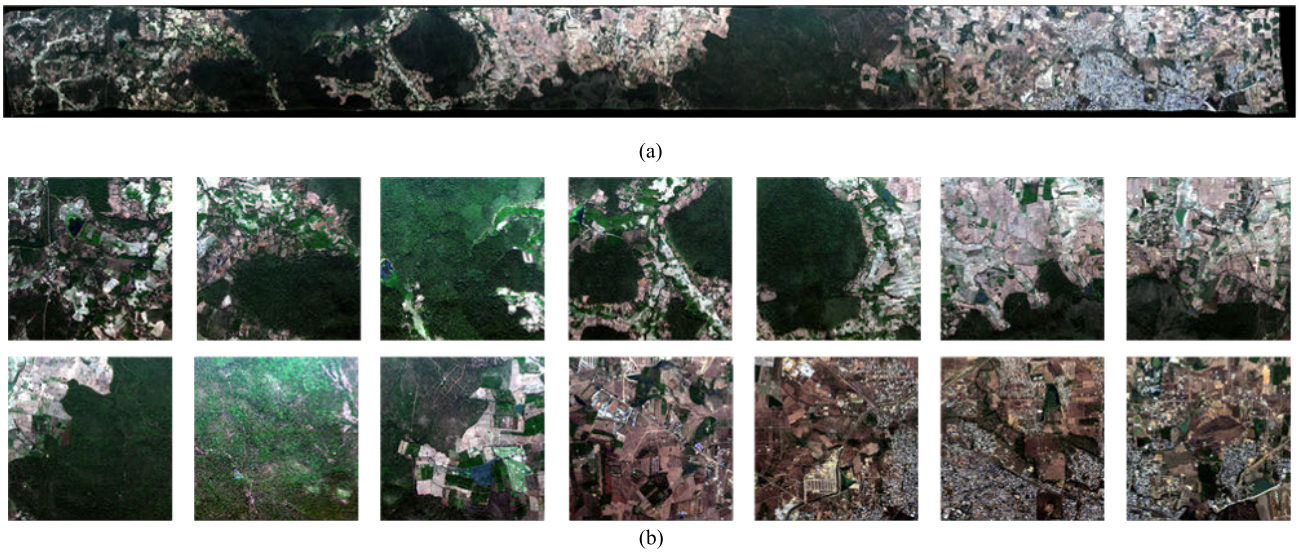


Fig. 11. (a) FC representation of the AVIRIS-NG image whose file identifier is ang20160101t072436. (b) FC representations of the tiles (512 × 512 pixels) extracted from the image.

for bad samples that are experienced in PRISMA images as discussed in [33].

Such analysis along with the possible strategies to mitigate the effects of those noise sources not modeled by (2) is the main goal of the ongoing research activity.

VI. CONCLUSION

In this work, we have proposed a new algorithm for XCO_2 retrieval from PRISMA hyperspectral data. The proposed algorithm is based on the RTM and, exploiting the estimates of the column H_2O and the spectral reflectance, reformulates the XCO_2 estimation problem in a linear form. This form makes it possible to tackle the problem by means of a computationally efficient algorithm based on the MF. The algorithm performance has been evaluated on a large simulated dataset of PRISMA-like images by testing the various hypotheses made in the derivation of the method. The analysis has included the comparison with the CIBR algorithm that is commonly adopted for estimating XCO_2 from hyperspectral data. Results on simulated data have shown that the proposed solution provides better estimation accuracy than CIBR with a gain in terms of RMSE of about 10 ppm. Such a conclusion has been confirmed by an example on real PRISMA data where the estimates provided by RTM-MF and those obtained by CIBR have been compared with OCO-2 data. It is worth noting that the conducted analysis has confirmed the limitations of hyperspectral imaging sensors in the estimation of XCO_2. The obtained errors are above those (in the order of few ppm) attained by processing data from very high spectral resolution sensors specifically designed for XCO_2 estimation (e.g., OCO-2 and OCO-3). However, the PRISMA mission is the precursor of a series

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of recently launched or forthcoming spaceborne spectroscopy missions with potential for CO_2 mapping. The availability of estimation algorithms specifically conceived for hyperspectral imaging sensors is a prerequite for the exploitation of the high spatial and temporal resolution observations that are or will be made available from the spaceborne missions mentioned above.

We would like to emphasize again that the analysis presented in this work does not include the effects of the spectral calibration errors that persist after smile and keystone noise correction, nor it does consider the impact of noise sources such as instrumental fixed pattern noise and bad samples, which are frequently encountered in real hyperspectral images. The analysis of those effects, together with the possible solutions for their mitigation, will be one of the objectives of our research activity in the near future.

APPENDIX

In this section, we provide details about the simulated dataset. In particular, in Table III, we list the AVIRIS-NG images adopted in our simulation. For each image, we provide the following information: the file identifier, the sensing date, the nominal ground sampling distance (GSD), the number of lines and samples, and the number of tiles extracted having size of 512 × 512 pixels. The total number of image tiles is 100.

In Fig. 11, for one of the considered AVIRIS-NG images (file identifier: ang20160101t072436), we show the FC representations of the entire image and the extracted tiles.

ACKNOWLEDGMENT

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Nicola Acito (Member, IEEE) received the Laurea degree (cum laude) in telecommunication engineering and the Ph.D. degree in "methods and technologies for environmental monitoring" from the University of Pisa, Pisa, Italy, in 2001 and 2005, respectively.

From November 2004 to October 2008, he was a Temporary Researcher with the Department of Information Engineering, University of Pisa. From March 2009 to December 2020, he was a Researcher/Lecturer with the Italian Navy Academy, Livorno, Italy. Since January 2021, he has been an Associate Professor with the University of Pisa. His research interests include multidimensional signal and image processing in remote sensing applications, deep learning, and machine learning techniques for remotely sensed data. His current activity has been focused on atmospheric compensation and super-resolution of hyperspectral images.

Dr. Acito is a reviewer of many international journals, i.e., IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON SIGNAL PROCESSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, and IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS.



M. Diani (Member, IEEE) received the Laurea degree (cum laude) in electronic engineering from the University of Pisa, Pisa, Italy, in 1988.

He is currently a Full Professor with the Italian Naval Academy, Livorno, Italy, where he teaches "signal processing" and "telecommunications theory." He also teaches "remote sensing" with the University of Pisa. His main research fields are image and signal processing with application to remote sensing. In this framework, his works have covered different topics, such as data fusion, signal processing in imaging radars, image classification and segmentation, and object detection. His recent activities have focused on the development of new algorithms for object detection and tracking in infrared image sequences and on target detection and recognition in multi/hyperspectral images.

Prof. Diani is an Associate Editor of the *Journal of Applied Remote Sensing* (JARS) and the *Journal of Real Time Image Processing* (JRTIP). He has served as a reviewer for many journals in the fields of remote sensing and image/signal processing, such as IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON SIGNAL PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and *Applied Optics*. He is a member of SPIE.

M. Alibani (Member, IEEE) received the Laurea degree in automation and robotics engineering and the Ph.D. degree in "advanced control and optimization for hybrid ROV/AUV vehicles" from the University of Pisa, Pisa, Italy, in 2017 and 2021, respectively.

He is a Research Assistant with the University of Pisa. His research interests include autonomous navigation systems for drone and UAV, multidimensional data processing, and data fusion.



G. Corsini (Life Member, IEEE) received the Laurea degree in electronic engineering from the University of Pisa, Pisa, Italy, in 1979.

He is currently a Full Professor with the Department of Information Engineering, University of Pisa, where he teaches "image processing." His main research fields include signal detection and processing with an emphasis on image and multidimensional signal analysis in remote sensing applications. His research interests include object detection and parameter estimation from remotely sensed data with particular emphasis on hyperspectral and multispectral images.

Prof. Corsini has been a member of technical committees at international conferences. He is a reviewer of the most important international journals in this field, i.e., IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON SIGNAL PROCESSING, IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS, and *The International Journal on Remote Sensing*.

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